

Ecological drivers and phylogenetic patterns of leaf minimum conductance variability in vascular plants

Santiago Trueba^{1,2} , Régis Burlett² , Xavier Bouteiller² , Thibaut Burneau², Guillaume Forget² , Maximilien Larter² , Daniela N'Do² , Camille Ziegler²  and Sylvain Delzon² 

¹AMAP, Univ Montpellier, IRD, CIRAD, CNRS, INRAE, Montpellier, 34398, France; ²University of Bordeaux, INRAE, UMR BIOGECO, Pessac, 33615, France

Author for correspondence:

Santiago Trueba

Email: santiago.trueba@ird.fr

Received: 2 December 2025

Accepted: 18 March 2026

New Phytologist (2026)

doi: 10.1111/nph.71166

Key words: angiosperms, conifers, cuticular conductance, drought, phenology, residual conductance, stomatal conductance, stomatal density.

Summary

- Stomatal closure prevents significant water losses during drought events. Yet, leaves are not perfectly hermetic and dehydration ensues through residual water losses, known as minimum conductance ($g_{\min}$), which is highly relevant since it informs on the water depletion dynamics under stress.
- We measured $g_{\min}$ on 101 species spanning phylogenetic and ecological diversities, from ferns to flowering plants. Sampling also included different growth forms and life cycles from annual herbs to longevous trees. We used stomatal measurements to estimate operational ($g_{\text{th op}}$) and maximum ($g_{\text{th max}}$) conductances.
- Minimum water conductance is highly variable across species, and $g_{\min}$ shows a weak phylogenetic signal across vascular plants. Annual herbaceous plants have greater water loss than woody plants, and deciduous species showed higher $g_{\min}$ rates than evergreen species. Relationships of $g_{\min}$ with $g_{\text{th op}}$ and $g_{\text{th max}}$ were weak, revealing the lack of a clear tradeoff between maximum potential conductance efficiency and water retention. We found reduced water expenses in species occupying hotter and more seasonal environments.
- This study integrates $g_{\min}$ in leaf economics, where long-lived leaves show higher capabilities to retain water under stress. We provide important information in evolutionary physiology to understand the water loss dynamics under drought stress across clades of vascular plants.

Introduction

Important innovations have evolved in terrestrial plants, allowing their establishment in more arid environments. Stomata, formed of an ostiole delimited by two guard cells, are amongst the key innovations that appeared in terrestrial embryophytes, allowing for the control of gas exchanges and water loss (Franks & Farquhar, 2007; Franks & Beerling, 2009). Along with stomata, the plant cuticle, a thin hydrophobic layer of lipidic monomers produced by the epidermal cells on the surface of the aerial organs (Riederer & Schreiber, 2001; Samuels *et al.*, 2008), also represents a major adaptation in response to drought in embryophytes by preventing excessive water loss. Understanding these past adaptive mechanisms is imperative in the context of climatic changes with increasing evaporative demands (Gebrechorkos *et al.*, 2025) already affecting plant hydraulic integrity, along with impacts on the geographical distribution of species, their phenology, productivity and survival (Allen *et al.*, 2010; Hammond *et al.*, 2022). In this context, differences in stomatal behavior as a response to drought stress across extant lineages and plant communities are an important axis of variation in plant ecological strategies (Brodribb & McAdam, 2017). When water becomes scarce, the survival of plants depends on a prompt

stomatal closure, which limits water loss and abrupt drops in water potential (Martin-StPaul *et al.*, 2017), thus delaying hydraulic dysfunction and plant desiccation, which can lead to cell damage and mortality (Mantova *et al.*, 2023).

Despite the effective control of water loss through stomatal closure, plants keep losing water at a significantly diminished rate (Duursma *et al.*, 2019). Such small rates of water loss can be exacerbated under the current trends of increasing vapor pressure deficit (VPD), which is the main driving force of plant evaporative demand (Grossiord *et al.*, 2020), bringing impacts on plant physiological processes and productivity (J. López *et al.*, 2021). Water loss after stomatal closure occurs through both stomatal leaks and through the cuticle, which, despite its hydrophobic nature, represents a second possible route for transpiration (Duursma *et al.*, 2019). Such post-stomatal closure water losses can be quantified by measuring the minimum leaf conductance to water vapor ($g_{\min}$; $\text{mmol m}^{-2} \text{s}^{-1}$). It has been recently shown that residual leaf conductance is highly variable over the course of dehydration, and that $g_{\min}$ has important impacts on the estimation of plant drought survival, with plants with higher $g_{\min}$ values being more susceptible to hydraulic failure, causing irreversible losses in water transport and supply to living tissues (Burlett *et al.*, 2025). Mechanistically, conductance rates of $g_{\min}$ play a

major role at regulating the pace at which plants cross their hydraulic safety margin (Petek-Petrik *et al.*, 2023), which is highly relevant given that the time to reach damaging water potentials is strongly related to mortality risk (Trueba *et al.*, 2024). Consequently, $g_{\min}$ holds an important value in process-based models using hydraulic traits for their parameterization, and attempting to predict plant mortality under drought (Cochard *et al.*, 2021; Ruffault *et al.*, 2022).

Given its relevance as a driver of water loss under stress, minimum conductance measurements in ecophysiological studies have experienced a big leap, with many studies exploring its variability within various ecosystems (e.g. Schuster *et al.*, 2016; Bueno *et al.*, 2019; Slot *et al.*, 2021; Schmitt & Boisseaux, 2023; Boisseaux *et al.*, 2025). A meta-analysis on cuticular transpiration across 160 species from various environments showed a lack of conductance variation across species with different growth forms and phenologies, suggesting a lack of ecological adaptation in cuticular water losses (Schuster *et al.*, 2017). Further, another systematic review on 221 species suggested that $g_{\min}$ variability was poorly related to species clades, climate or leaf type (Duursma *et al.*, 2019). Although these are considerable contributions, both meta-analyses gathered data from studies using various techniques and measurement protocols. Given that methodological differences can lead to strong variation in conductance estimations and induce major biases in determining plant drought survival (Burllett *et al.*, 2025), measurements using a single, stable protocol are needed to better grasp the variation of $g_{\min}$ across species with different growth forms and phenologies.

Complexity of stomatal behavior in seed plants has evolved from a common ancestor with a simple passive stomatal control existing in ferns and lycophytes to the emergence of active stomatal regulation in angiosperms and gymnosperm clades, which consequently enhanced water use efficiency (Brodribb & McAdam, 2011, 2017; McAdam & Brodribb, 2012; Yang *et al.*, 2021). In this context, it is highly relevant to assess whether this evolutionary history also contributed to lower water loss rates under severe stress in vascular plants. Moreover, given that trait values result from the interplay of ecological and evolutionary processes, studying the phylogenetic signal can provide valuable information regarding the tendency of related species to exhibit similar trait values (Münkemüller *et al.*, 2012). Consequently, we need to explore the distribution of minimum conductance onto a phylogeny of vascular plants, and analyze its phylogenetic signal to unveil evolutionary patterns of water loss after stomatal closure.

Stomata are responsible for the high transpiration rates needed for CO₂ fixation and plant growth in well-watered conditions (Wong *et al.*, 1979), and part of the residual water losses during prolonged drought (Brodribb *et al.*, 2014). Therefore, there is a potential tradeoff between maximizing productivity via high maximum transpiration rates ($g_{\text{th max}}$; mmol m⁻² s⁻¹) and reducing $g_{\min}$. Maximum transpiration rates through $g_{\text{th max}}$ can be estimated with stomatal anatomical traits (Franks & Beerling, 2009; Sack & Buckley, 2016). Consequently, large stomatal openings, a high stomatal index, the investment of stomata

relative to epidermal cells, or high stomatal densities can be correlated with higher maxima in transpiration rates (Ochoa *et al.*, 2024). Incomplete or imperfect stomatal closure is a potential pathway of water loss driving minimum leaf conductance. Previous studies have directly or indirectly assessed this, with evidence of stomatal leakiness from various unrelated species such as *Hedera helix* (Šantrůček *et al.*, 2004), *Sorghum bicolor* genotypes (Muchow & Sinclair, 1989), or across conifer species (Brodribb *et al.*, 2014). We could therefore expect coordination between $g_{\min}$ and $g_{\text{th max}}$ with a potential tradeoff between both sides of the conductance spectrum. From an evolutionary perspective, species with high leaf water conductance efficiency would be at a risk and dehydrate faster during stress events, so we would further expect this tradeoff to be reflected in species distributions.

Direct linkages of minimum leaf conductance rates with environmental variables can be highly valuable to underline their adaptive value. It has been shown that $g_{\min}$ is related to the period of the year with the most intense water stress in 42 conifer species (Brodribb *et al.*, 2014). In this line, a study on the intra-variability of two *Hackea* species from xeric habitats across Australia suggested that decreasing $g_{\min}$ allows one to withstand the impact of increasing aridity (R. López *et al.*, 2021). To extend these observations, we need further investigation on the potential adaptive role of $g_{\min}$ to withstand aridity using a wider set of species from various environmental envelopes. In this study, we measure the dynamics of water loss on both ends of its spectrum, from the minimum to the maximum rates of water loss in 101 species of vascular plants, from ferns to angiosperms. For the first time at this scale, we measured $g_{\min}$ using a uniform methodology, with a special emphasis on using water status thresholds, as recommended by a recent protocol to maximize comparability between species (Burllett *et al.*, 2025). We explore the variability of $g_{\min}$ across species and phenology types. Moreover, we explore potential coordination of $g_{\min}$ with stomatal-derived traits and assess how minimum conductance values relate to the environmental distribution of species. We specifically tested the following hypotheses: (1) $g_{\min}$ has a weak phylogenetic signal, as it can be a convergent character strongly selected by environmental pressure, with species developing higher water losses in more permissive habitats; (2) higher $g_{\min}$ variability in angiosperms relative to gymnosperms and ferns, reflecting the higher diversity of angiosperm ecological strategies; (3) a tradeoff between $g_{\min}$ and $g_{\text{th max}}$, driven by stomatal traits; (4) water stress drives minimum conductance variation with low $g_{\min}$ in arid environments as a mechanism allowing one to avoid rapid dehydration.

Materials and Methods

Species sampling and plant material

We studied 101 vascular plant species belonging to 57 families and 40 orders within both spermatophytes and polypodiophytes (Fig. 1). We standardized spelling and nomenclature of species using the world flora online (WFO v.1.1; <https://wfoplantlist.org/>). Sampling was particularly extensive in angiosperms with

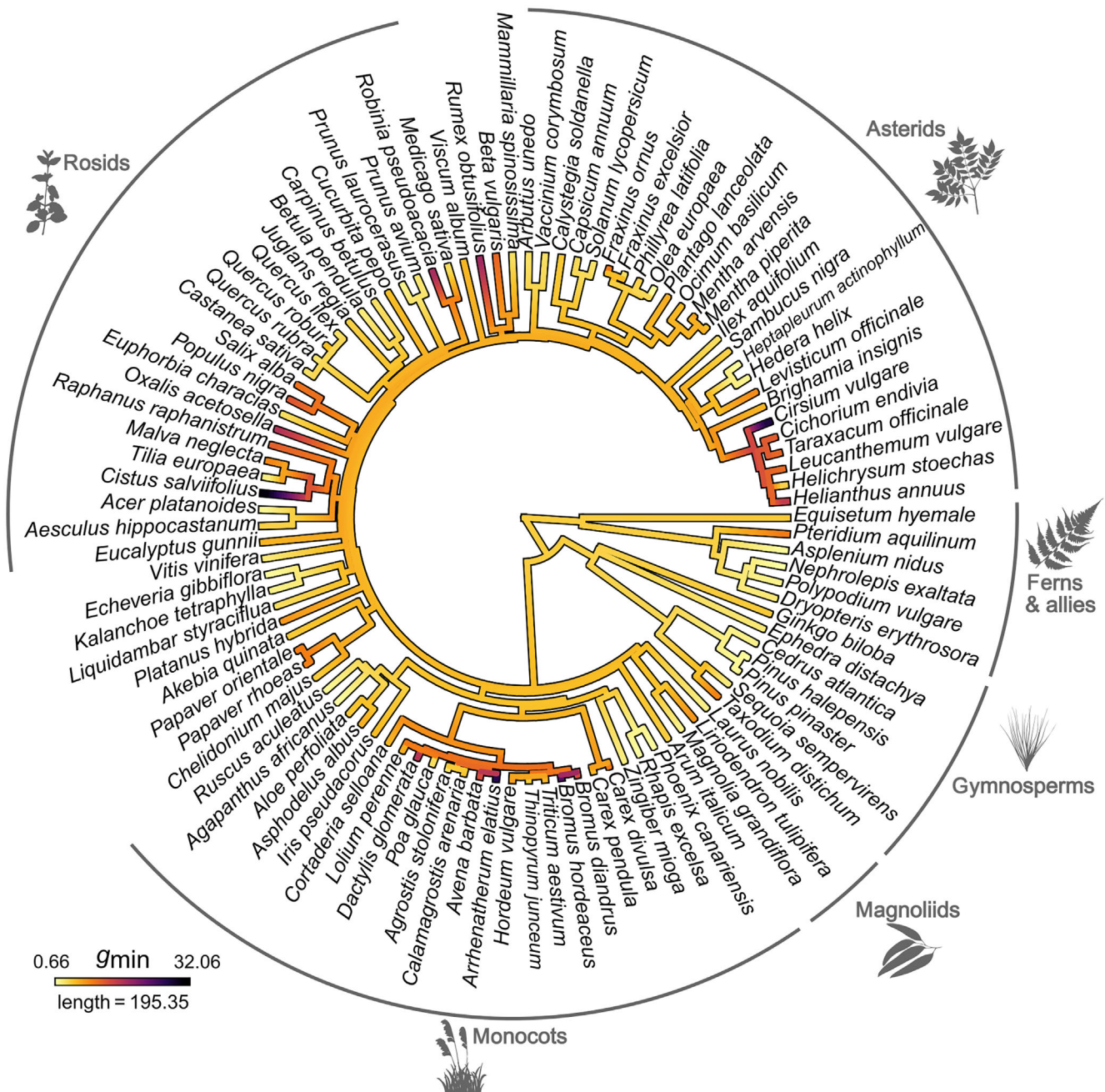


Fig. 1 Minimum conductance to water vapor after stomatal closure ($g_{\min}$) across vascular plant species. Conductance is mapped on a phylogenetic tree of 101 species. The color gradient along the branches of the tree indicates lower (light) and higher (dark) values of $g_{\min}$. A scale bar (in My) is included. Major clades of sampled vascular plants are indicated. The species not included in any group are all eudicot angiosperms, which also include the ensemble of asterids + rosids. Reconstructed $g_{\min}$ values across the nodes of the phylogeny are available as Supporting Information (Fig. S2; Table S1).

88 species from major clades of flowering plants (e.g. asterids, rosids, and monocots), and completed with 7 gymnosperms and 6 ferns and allies (Fig. 1). A large diversity of growth forms is represented with strict herbs, woody shrubs and vines, as well as succulent plants (e.g. *Aloe perfoliata* L., *Brighamia insignis* A. Gray and *Mammillaria spinosissima* Lem.) and very tall trees (e.g.

Sequoia sempervirens (D. Don) Endl.). Our sampling also represents plants with particular ecologies such as *Calamagrostis arenaria* (L.) Roth, which grows in coastal sand dunes, or the hemiparasitic habit of the mistletoe *Viscum album* L. Moreover, we also included many crops and cultivated plants, among these are basil (*Ocimum basilicum* L.), beet (*Beta vulgaris* L.), endive

(*Cichorium endivia* L.), myoga ginger (*Zingiber mioga* (Thunb.) Roscoe), pepper (*Capsicum annuum* L.), pumpkin (*Cucurbita pepo* L.), sunflower (*Helianthus annuus* L.), tomato (*Solanum lycopersicum* L.) and wheat (*Triticum aestivum* L.). Plants were mostly obtained within the plant collection of the campus of the University of Bordeaux (Pessac, France), as well as in natural areas in the surroundings of Bordeaux, France. Additional species, particularly crops, ornamental and tropical species were purchased from plant nurseries.

We carried out 49 different measurement campaigns between 2020 and 2025, prioritizing collections of plant material during mid to late spring to avoid peaks of drought and heat stress. We collected fully developed and undamaged leaves of mature individuals for all measurements, and cultivated, potted plants were acclimated in a growth chamber for a minimum of 30 d. We measured an average of eight leaves per species, belonging to at least three different individuals. In some cases, we rejected bad quality dehydration curves, with the final dataset including at least four samples per species after data curation. For most species, we measured single, detached leaves, or the equivalent functional photosynthetic unit such as cladodes (i.e. flat shoots with the appearance of leaves) in *Ruscus aculeatus* L., or the photosynthetic stems in *Equisetum hyemale* L. or *Mammillaria spinosissima*, which bears spines instead of true leaves. We distinguished plants with four main different phenology types (annual herb, perennial herb, woody deciduous and woody evergreen). Within sampled angiosperms 55.7% of species are herbaceous (24 species annual and 25 species perennial) and 44.3% are woody (21 species deciduous and 18 species evergreen). Sampled woody gymnosperms are mainly evergreen with only two out of seven species being deciduous.

Experimental dehydration and minimum conductance measurements

After *in-situ* collections or harvesting, samples were placed in dark plastic bags with wet paper towels to avoid dehydration and transported to the lab within 2 h after collection. Leafy shoots were recut underwater by three nodes, placed with cut ends in water-filled buckets, covered in plastic to achieve vapor-saturation, and rehydrated overnight for at least 12 h following standard rehydration protocols (Trueba *et al.*, 2019). The day after collection leaves were excised from the shoots and were immediately weighed with a 4-digit balance (Pioneer, Ohaus, Parsippany, NJ, USA) to obtain the water-saturated turgid mass (M_i in g). Leaves were then scanned using a flatbed scanner (v.850 pro, Epson, Japan) and images were analyzed with IMAGEJ v.1.54f (Schneider *et al.*, 2012) to calculate projected leaf area (A_{leaf} in m^2). After initial weight and area estimations, leaves were tagged and the base of the petiole was sealed with paraffin wax to prevent direct dehydration from the cut-end. Similar procedures were employed in particular samples such as leafless stems (see previous section), but the developed area was instead employed using dimensional measurements from caliper measurements assuming the surface area of a cylinder as

$A_{leaf} = 2\pi r^2 + 2\pi rh$, where r and h are the radius and the height of the sample, respectively.

Dehydration dynamics of detached leaves were estimated using a *droughtbox* device (Billon *et al.*, 2020) at the *Phenobois* platform (INRAE, Pessac, France). Monitoring of weight loss was obtained by weighing every 5 min with 24 load cells (3139_0; Phidgets Inc., Calgary, AB, Canada) enclosed in a 1200 l growth chamber (Fitoclima 1200, Aralab, Portugal) allowing stable conditions during dehydration. A detailed description of the custom setup and software of the *droughtbox* device is provided in Burlett *et al.* (2025). Temperature and relative humidity (RH) in the chamber were set to 25°C and 60%, respectively, resulting in an air VPD of *c.* 1.27 kPa. Samples were illuminated during measurements from the top and from the bottom with a Photon Flux Density of 400 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Illumination during the experiment was set up to ensure that stomatal closure resulted from dehydration and not from a lack of light stimulation. Samples were left to dehydrate for variable time lengths depending on the pace of water loss but in most cases, experiments lasted up to 7 d to ensure significant dehydration. At the end of the measurement, leaves were put in an oven at 65°C for 72 h and dry mass (M_d) was measured. The relative water content (RWC) was then computed for each mass value recorded in the *droughtbox* (fresh mass; M_f) during the experiments using Eqn 1:

$$RWC = 100 \times \left(\frac{M_f - M_d}{M_t - M_d} \right) \quad \text{Eqn 1}$$

Minimum conductance was computed for each sample as the rate of water loss divided by its driving force (VPD) using Eqn 2:

$$g_{\min} = \frac{dw/dt}{M_{H_2O} \times A_{leaf}} \times \frac{P_{\text{atm}}}{VPD} \quad \text{Eqn 2}$$

where dw/dt is the negative slope of the weight–time relationship from the dehydration curve of the experiment, M_{H_2O} is the molecular weight of water (18.01 g mol^{-1}), A_{leaf} is the projected leaf area (in m^2), P_{atm} is the atmospheric pressure in the chamber (set to *c.* 101.9 kPa), and VPD is the vapor pressure deficit between the inner leaf and the air inside the chamber (in kPa). Computation of minimum conductance ($g_{\min}$) was carried out with the Python console-based software GMINCOMPUTATION v.2.0 (University of Bordeaux; <https://gitub.u-bordeaux.fr/phenobois>). We used continuous RWC estimations to measure the rates of water loss between the boundaries of 80% and 50% RWC, and to estimate time_{RWC80} and time_{RWC50} , the elapsed time to reach RWC thresholds. Such boundaries were set as previous research suggests that these are close to the points of stomatal closure and irreversible dehydration, respectively (Martin-StPaul *et al.*, 2017; Trueba *et al.*, 2019). A recent study has shown that $g_{\min}$ at RWC_{80-50} provides reliable conductance estimations close to $g_{\min}$ at stomatal closure (Burlett *et al.*, 2025). Given the lack of knowledge of physiological thresholds indicating stomatal closure, such as g_s declines or turgor loss point (Brodribb *et al.*, 2003), for most of the studied species, such an approach provided a reliable proxy to

seize water loss after stomatal closure and compare dehydration dynamics across a large variety of species.

Stomatal anatomy, theoretical conductance estimations, and leaf traits

Measurements of stomatal anatomy were carried out on a subset of 87 species. Missing observations on the 14 remaining species resulted from a lack of material or difficulty of stomatal observations mainly due to stomatal crypts and high trichome presence. Stomatal trait measurements were carried out on three leaves per species that were at proximity from the leaves used in the experimental dehydration for conductance estimations. Nail varnish impressions were carried out on both leaf surfaces, and image acquisition was done with an optical microscope (Leica DM2500; Leica Microsystems, Wetzlar, Germany). Ten times magnification images were used to estimate stomatal density (SD, in stomata mm^{-2}), while 40 $\times$ images were used to carry out stomatal trait measurements, including stomatal surface (A_{sto} , in μm^2), on 12 randomly selected stomata per leaf. Other stomatal measurements included stomatal length (L , in m) and width (W , in m), guard cell width (W_g , in m), stomatal pore surface (A_p , in m^2), and epidermal cell surface (A_{epi} , in m^2). All measurements were carried out with IMAGEJ v.1.54f. We used measurements of g_{min} as a baseline conductance, along with stomatal measurements, to estimate different rates of leaf water conductance based on recently established equations (Ochoa *et al.*, 2024). We estimated $g_{\text{th op}}$, the leaf operational conductance under high irradiance, using a pore with mid-aperture ($\alpha = 0.5$), and $g_{\text{th max}}$, the leaf maximum diffusional conductance assuming all stomata fully open ($\alpha = 1$), using Eqn 3:

$$g_{\text{th}} = \frac{D_{\text{wa}}\pi\alpha^2\mu^2\text{SD}_U A_{\text{sto},U}}{V_{\text{ma}}(4j + \pi\sqrt{\alpha})(\chi' + \alpha\mu\sqrt{A_{\text{sto},U}})} + \frac{D_{\text{wa}}\pi\alpha^2\mu^2\text{SD}_L A_{\text{sto},L}}{V_{\text{ma}}(4j + \pi\sqrt{\alpha})(\chi' + \alpha\mu\sqrt{A_{\text{sto},L}})} + \left(1 - \frac{\pi}{4}\alpha\mu^2\text{SD}_U A_{\text{sto},U}\right)g_{\text{ec}} + \left(1 - \frac{\pi}{4}\alpha\mu^2\text{SD}_L A_{\text{sto},L}\right)g_{\text{ec}} \quad \text{Eqn 3}$$

where SD_U and SD_L are stomatal densities (stomata m^{-2}) for adaxial and abaxial surfaces, respectively; $A_{\text{sto},U}$ and $A_{\text{sto},L}$ are stomatal area (m^2) for adaxial and abaxial surfaces, respectively. χ' is the mean free path in air (i.e. the mean distance a gas molecule travels before colliding with another, m); D_{wa} is the binary diffusivity of water vapor in air ($\text{m}^2 \text{s}^{-1}$); V_{ma} is the molar volume of air ($\text{m}^3 \text{mol}^{-1}$); μ and j are the dimensionless scaling factors ($\mu = 1/\sqrt{2}$ and $j = 0.5$ for non-grasses with kidney bean-shaped guard cells, and $\mu = \sqrt{2}$ and $j = 0.125$ for grasses with dumbbell shaped guard cells); g_{ec} is the minimum conductance ($\text{mmol m}^{-2} \text{s}^{-1}$) divided by two surfaces and estimated between 80% and 50% of RWC. Additional estimations of leaf maximal conductance ($g_{\text{th maxFB}}$; in $\text{mmol m}^{-2} \text{s}^{-1}$) were also performed using previously established estimations (Franks & Beerling, 2009) of conductance maxima as a function of stomatal features

(Supporting Information Methods S1). Along with the previous stomata-related parameters used to estimate leaf conductance, we used $A_{\text{epi},U}$ and $A_{\text{epi},L}$, the epidermal cell area (in m^2) for adaxial and abaxial surfaces, respectively, to estimate the stomatal index (SI; unitless), using previously established calculations (Sack & Buckley, 2016) with Eqn 4:

$$\text{SD}_{\text{tot}} = \text{SD}_L + \text{SD}_U$$

$$A_{\text{epi}} = \frac{A_{\text{epi},L} + A_{\text{epi},U}}{2}$$

$$A_{\text{sto}} = \frac{A_{\text{sto},L} + A_{\text{sto},U}}{2}$$

$$\text{SI} = \frac{A_{\text{epi}}}{\left(\frac{1}{\text{SD}_{\text{tot}}} - A_{\text{sto}} + A_{\text{epi}}\right)} \quad \text{Eqn 4}$$

Finally, we used $L_{\text{sto},U}$ and $L_{\text{sto},L}$, the stomatal length (m) for adaxial and abaxial surfaces, respectively, to calculate the stomata pore index (SPI; unitless) based on previous studies (Liu *et al.*, 2023) with Eqn 5:

$$L_{\text{sto}} = \frac{L_{\text{sto},L} + L_{\text{sto},U}}{2}$$

$$\text{SPI} = \text{SD}_{\text{tot}} L_{\text{sto}}^2 \quad \text{Eqn 5}$$

Along with the stomatal-derived traits, we also calculated the degree of succulence (Delf, 1912) as the water content per unit area of surface as $\text{suc} = (M_t - M_d)/A_{\text{leaf}}$ to explore potential linkages of succulence with water retention via decreased minimum conductance. Additionally, we estimated leaf mass per area as $\text{LMA} = M_d/A_{\text{leaf}}$ with the hypothesis that leaf mass investment per unit area could be negatively related to g_{min} across species as a syndrome of leaf longevity implicating water retention.

Species distribution and climate variables

We extracted occurrence data for all species in the g_{min} dataset from the Global Biodiversity Information Facility (GBIF) database using the *rgbif* interface. We matched exact or accepted nomenclatures, and filtered only records from preserved specimens and natural locations. We used the 'clean_coordinates' function of the COORDINATECLEANER package (Zizka *et al.*, 2019) to remove points with coordinates placed as default locations (such as country capitals, country centroids, large biodiversity institution headquarters), in oceans, and outliers (based on the distance from other points for the same species). The remaining occurrence points were then filtered to remove any falling outside the natural ranges of species. To this end, we mapped the remaining points using the LEAFLET and LEAFLET.EXTRAS packages (Cheng *et al.*, 2024; Gatscha *et al.*, 2024) to interface with leaflet, an open-source JavaScript library for interactive maps (<https://leafletjs.com/index.html>). We manually drew one or several polygons around all occurrences within the natural range of each

species. For crops, we used the ancestral range of the species. We then selected GBIF occurrences within each species' range as defined by the resulting polygons using the 'sf' package (Pebesma, 2018). This reduced the median number of occurrences per species by *c.* 50%, from *c.* 100 000 to 50 000 points per species. Four species had more than a million occurrence points (*Dactylis glomerata* L., *Fraxinus excelsior* L., *Plantago lanceolata* L., *Quercus robur* L.) and five had fewer than 100 (*Brighamia insignis*, *Calystegia soldanella* (L.) R.Br., *Kalanchoe tetraphylla* H.Perrier, *Mammillaria spinosissima*, *Platanus hybrida* Brot.).

We downloaded the WORLDCLIM v.2.1 (Fick & Hijmans, 2017) climate layers at 2.5-min resolution and the Global Aridity Index and Potential Evapotranspiration Climate Database v.2 (Trabucco & Zomer, 2018). We then extracted climate values at all occurrences using the TERRA package (Hijmans, 2025). We summarized the climate data distribution for each species by computing the mean, median, 5th and 95th percentiles. Because climate variables can be highly correlated, we selected six WorldClim variables *a priori* that could explain the variation of $g_{\min}$ and $g_{\text{th max}}$. These were the mean annual temperature and precipitation (BIO1 and BIO12, respectively), temperature and precipitation seasonality (BIO4 and BIO15, respectively), and finally the variables that capture the climate extremes of temperature (BIO5; maximum temperature of the warmest month) and precipitation (BIO17; precipitation of the driest quarter).

We evaluated the impact of removing exotic occurrences on estimating species climate envelopes by looking at the correlation between climate variables across species with and without the exotic occurrences. Overall, the correlations were strong and significant ($R^2 > 0.7$; $P < 0.001$), with most of the points falling very close to the 1 : 1 line. However, there are a number of outliers, mostly crops (e.g. tomato, basil, sunflower, barley, and pumpkin), some widely planted tree species with narrow natural ranges (e.g. *Taxodium distichum* (L.) Rich., *Ginkgo biloba* L., *Aesculus hippocastanum* L., and *Cedrus atlantica* (Endl.) Manetti ex Carrière), and some species with large introduced or invasive ranges (e.g. *Robinia pseudoacacia* L. and *Quercus rubra* L.).

Data analyses

We established phylogenetic relationships across the studied species (Fig. 1) using as backbone the mega-tree 'GBOTB.extended.tre' included in the package V.PHYLOMAKER (Jin & Qian, 2019), which combines dated phylogenies for spermatophytes (Smith & Brown, 2018) and pteridophytes (Zanne *et al.*, 2013). To investigate the influence of evolutionary history on trait expression, we assessed the degree of trait similarity in closely related species using the phylogenetic signal. We used Blomberg's *K* (Blomberg *et al.*, 2003) to test for phylogenetic signal by comparing *K* for the observed trait phylogenetic distribution to a null distribution obtained with 1000 trait permutations across the phylogeny. We additionally used Pagel's λ (Pagel, 1999) with significance assessed using a likelihood ratio test. A phylogenetic signal value close to zero indicates phylogenetic independence, while a value of one indicates that the trait

in close relatives is more similar than expected under a Brownian motion model of trait evolution. We mapped leaf conductance values on the vascular plant phylogeny and estimated the maximum likelihood (ML) ancestral character states onto the nodes of the phylogeny. Both phylogenetic signal and ML ancestral states were implemented with the package PHYTOOLS (Revell, 2012).

Assumptions of residual homogeneity and normality were tested before analyses. Comparisons of structural and physiological traits across groups were performed on species mean values using one-way ANOVA with *post hoc* Tukey's honest significant differences using 95% confidence intervals. Further comparisons were performed using an independent two-sample *t*-test when appropriate. Given that trait variability can be influenced by various nested scales (Messier *et al.*, 2010), we explored the contribution of taxonomic and ecologic factors to $g_{\min}$ variability. We partitioned the variance of $g_{\min}$ across five levels (group, order, family, growth form, and phenology) with a general linear model, sequentially calculating the variance explained at each hierarchical level.

Additionally, we employed phylogenetic Bayesian linear mixed-effects modeling (BLMM) to quantify the relative importance of phylogeny, along with phenology and growth form groupings, in explaining $g_{\min}$ variability. BLMM were implemented in the probabilistic programming language Stan (Carpenter *et al.*, 2017) using the BRMS package (Bürkner, 2017). The BLMM was fit with random intercepts for growth form and phenology, plus a phylogenetic random effect at the species level using a phylogenetic covariance matrix to estimate species covariation based on shared evolutionary history. The Hamiltonian Monte Carlo sampler achieved good convergence over 12 000 iterations. Convergence was assessed using the potential scale reduction factor (Rhat < 1.01 for all parameters), effective sample size ratios, and visual inspection of chain mixing and stationarity. We extracted the posterior distributions of variance components to display the percentage of total variance explained by phylogeny, growth form, phenology, and the residual error.

To estimate the strength of bi-variate relationships, we used linear regressions, along with Pearson's correlations. The relationships between water conductance rates and climate variables were assessed both on the full dataset and by separating species by their phenology. Additionally, we analyzed relationships between traits using phylogenetic independent contrasts (PIC) and phylogenetic generalized least-squares analyses (PGLS) with a lambda (λ) ML optimization to control for phylogenetic non-independence between related species (Felsenstein, 1985; Freckleton *et al.*, 2002). Trait data was species-averaged before PIC and PGLS analyses, which were carried out using the packages APE and CAPER (Paradis *et al.*, 2004; Orme *et al.*, 2018). A principal component analysis (PCA) was used to explore the covariation of leaf conductance and functional traits along with the distribution of phenology types. Given that the PCA can only be carried out on complete datasets, we coped missing stomata-related data for 14 species (13.9% of the PCA dataset) by imputing missing values using the MISSMDA package (Josse & Husson, 2016). All data manipulation and analyses were carried out using R 4.5.2 (R Core Team, 2020).

Results

Minimum leaf conductance variability in vascular plants

Across all measured samples we observed an average $g_{\min}$ between 80% and 50% RWC boundaries of $6.13 \pm 0.23 \text{ mmol m}^{-2} \text{ s}^{-1}$. We observed great interspecific variability in $g_{\min}$ ($F = 45.35$ (100); $P < 0.001$) with species mean conductance values varying *c.* 50-fold (Figs 1, S1); species with the highest and lowest minimum conductance were *Cistus salvifolius* L. and *Nephrolepis exaltata* (L.) Schott with values of 32.06 and $0.66 \text{ mmol m}^{-2} \text{ s}^{-1}$, respectively (Figs 1, S1). We found no evidence of phylogenetic signal in $g_{\min}$ (Fig. 1) with a low and non-significant Blomberg's $K = 0.10$ ($P = 0.60$), further confirmed with a low Pagel's $\lambda = 0.11$ ($P = 0.89$). Our findings suggest evolutionary convergence of species toward high values of $g_{\min}$ (Fig. 1). For instance, within flowering plants, the high $g_{\min}$ values recorded in extant species from Asteraceae (asterids), Cistaceae (rosids), and Poaceae (monocots) seem to have evolved from a common angiosperm ancestor with moderate conductance values (Fig. 1; Table S1; estimated ML reconstruction = $5.51 \text{ mmol m}^{-2} \text{ s}^{-1}$). This was also evident when comparing $g_{\min}$ across major clades of vascular plants (Fig. 2), with no significant differences between angiosperm clades, asterids, rosids, monocots and other non-monophyletic groups (including magnoliids). Ferns and allies (i.e. *Equisetum hyemale*) had intermediary conductance values similar to those of most of angiosperms and gymnosperms, which had the lower $g_{\min}$ values overall (Fig. 2). However, such differences in $g_{\min}$ were not evident when comparing species-averaged values across clades (Fig. 2; Table S2). A similar lack of phylogenetic signal was observed in LMA and succulence across the 101 sampled species, while A_{leaf} had a significant phylogenetic signal, suggesting similar leaf sizes within closely related species (Table 1).

Minimum leaf conductance variability across phenologies and clades

Differences in $g_{\min}$ across phenology types were significant (Fig. 3; $F = 48.74(3)$, $P < 0.001$). Annual herbs, which have the shortest life cycle, showed the highest rates of water loss through leaf $g_{\min}$ with an average value of $9.92 \pm 0.61 \text{ mmol m}^{-2} \text{ s}^{-1}$ followed by perennial herbs, which had significantly lower conductance rates ($6.67 \pm 0.42 \text{ mmol m}^{-2} \text{ s}^{-1}$). Interestingly, perennial herbs had similar $g_{\min}$ values than woody deciduous species, which had mean conductance values of $5.62 \pm 0.35 \text{ mmol m}^{-2} \text{ s}^{-1}$ that are significantly higher than those recorded for woody evergreen species ($2.35 \pm 0.13 \text{ mmol m}^{-2} \text{ s}^{-1}$). Annual herbs showed large variability across the $g_{\min}$ values spectrum, while woody evergreen samples showed lower variability and a very high concentration at low conductance values (Fig. 3). $g_{\min}$ variability was also significant when comparing species-averaged values across phenology types (Table S3). When considering woody angiosperms and gymnosperms independently, only angiosperm deciduous species had significantly higher $g_{\min}$ than their evergreen counterparts (Fig. S3). Hierarchical variance partitioning showed that phenology explained the largest proportion of variation in $g_{\min}$ (Fig. 4), followed by growth form, and group (i.e. angiosperms, ferns and allies, and gymnosperms). Family and order explained modest but non-negligible proportions, and we observed a very low residual variation (Fig. 4). A phylogenetic mixed model (BLMM) further comforted the importance of phenology as an important driver of $g_{\min}$ variability (Fig. S4), predominant over growth form or phylogeny. Phylogeny explained only 4.3% of total $g_{\min}$ variability, which could be expected given a negligible phylogenetic signal. The BLMM variance explained by phenology (12.3%) was, however, lower than the residual error fraction (77.6%), suggesting that an important fraction of

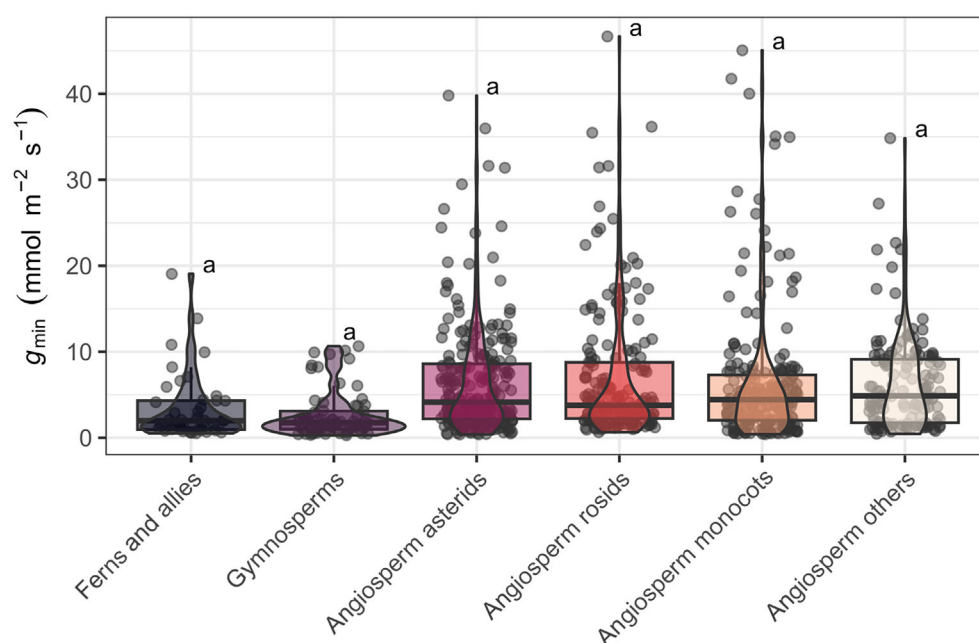


Fig. 2 Comparisons of minimum conductance ($g_{\min}$) values across clades of vascular plants. Major monophyletic groups of angiosperms such as asterids, rosids, and monocots are distinguished; other paraphyletic groups are included as angiosperm others. Other sampled species were classed as ferns & allies and gymnosperms. Distributions of conductance values using density curves are indicated; the width of each curve corresponds to the approximate frequency of data points in each region. Boxes and bars show the median, quartiles, and extreme values. Dots are individual data points. Letters indicate differences in $g_{\min}$ across plant clades within the species-level dataset.

Table 1 Synthesis of the studied conductance, stomatal and leaf traits.

Trait	Abbreviation	Unit	<i>n</i>	Mean ± SE	Min	Max	Bloomberg's <i>K</i>	Pagel's λ
Minimum leaf conductance	$g_{\min}$	$\text{mmol m}^{-2} \text{s}^{-1}$	101	6.37 ± 0.66	0.66	32.06	0.10	0.11
Maximum theoretical leaf conductance	$g_{\text{th max}}$	$\text{mmol m}^{-2} \text{s}^{-1}$	87	1750.02 ± 117.19	86.35	6214.38	0.18	0.63
Operational theoretical leaf conductance	$g_{\text{th op}}$	$\text{mmol m}^{-2} \text{s}^{-1}$	87	1080.93 ± 73.25	54.03	3770.39	0.19	0.65
Maximum theoretical leaf conductance, Franks & Beerling (2009)	$g_{\text{th max FB}}$	$\text{mmol m}^{-2} \text{s}^{-1}$	87	1421.39 ± 86.41	44.44	4891.36	0.18	0.41
Leaf area	A_{leaf}	cm^2	101	63.89 ± 4.88	6.62	281.51	0.22	0.82
Time to reach 80% RWC	$\text{time}_{\text{RWC}80}$	hours	101	16.30 ± 2.46	0.22	171.62	0.36	1
Time to reach 50% RWC	$\text{time}_{\text{RWC}50}$	hours	101	48.55 ± 8.28	2.49	600.92	0.23	0.00
Leaf mass per area	LMA	g m^{-2}	101	142.75 ± 10.56	26.12	629.60	0.16	0.48
Succulence	<i>suc</i>	g m^{-2}	101	377.93 ± 71.44	87.68	5155.11	0.23	0.00
Stomatal pore index	SPI	%	87	0.147 ± 0.010	0.004	0.507	0.13	0.52
Stomatal index	SI	%	87	0.190 ± 0.011	0.011	0.436	0.18	0.74
Stomatal surface	A_{stom}	μm^2	87	513.65 ± 52.55	30.08	2425.88	0.77	0.80
Stomatal density	SD	stomata mm^{-2}	87	199.40 ± 16.01	9.33	736.00	0.24	0.63

Species level mean ± SE along with extreme values across vascular plant species are included for each trait. Blomberg's *K* and Pagel's λ values in bold were significantly different from 0, indicating a significant ($P < 0.05$) phylogenetic signal. The number of species sampled for each trait measurements is specified. $g_{\min}$ corresponds to minimum conductance values measured between 50% and 80% relative water content.

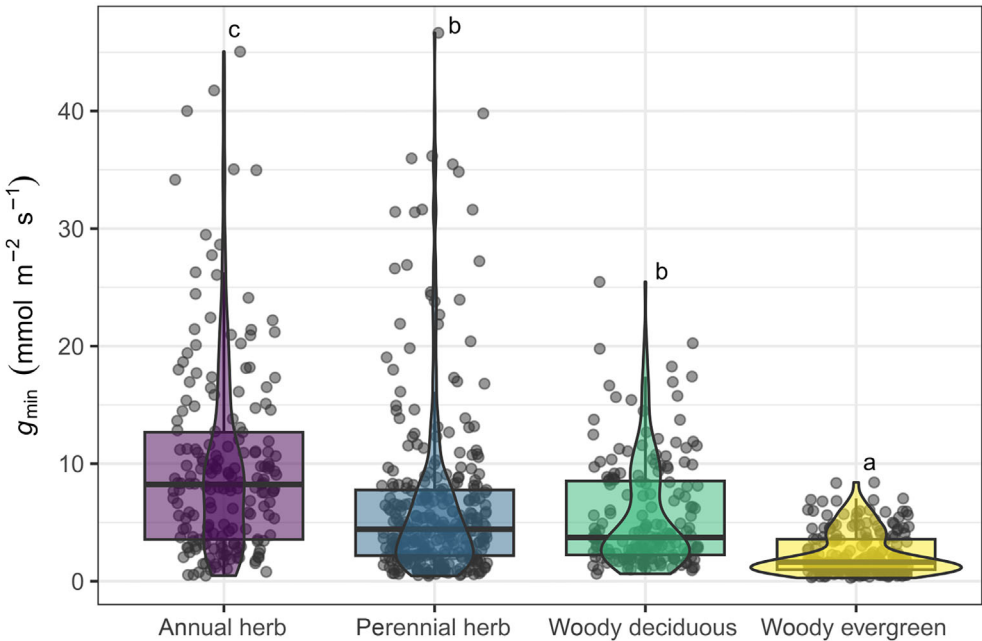


Fig. 3 Comparisons of minimum conductance ($g_{\min}$) values across phenologies in herbaceous and woody species. Herbaceous plants with different life cycles (annual vs perennial) and woody plants with different leaf habits (deciduous vs evergreen) are indicated. Distributions of conductance values using density curves are indicated; the width of each curve corresponds to the approximate frequency of data points in each region. Boxes and bars show the median, quartiles, and extreme values. Dots are individual data points. Letters indicate significant differences in $g_{\min}$ across phenologies within the species-level dataset. Further comparisons across phenologies within woody angiosperm and gymnosperm species are available as Supporting Information (Fig. S3).

minimum conductance variability might be explained by environmental or species-specific predictors not included in the model. Extensive comparisons between clades and phenologies for all measured traits from species-averaged values are available as Supporting Information (Tables S2, S3).

Stomatal traits and stomata-derived estimations of operational and maximum conductance

Across the subsample of 87 measured species for stomatal traits, 41 species were hypostomatous, presenting stomatal pores exclusively on the lower surface, and 46 species were amphistomatous,

with stomatal pores on both leaf surfaces. The SI, the proportion of stomata relative to epidermal cells, had a mean value of 0.19 ± 0.01 and varied from 0.01 to 0.43 in *Chelidonium majus* L. and *Capsicum annuum*, respectively. The estimation of the maximum leaf conductance ($g_{\text{th max}}$), using the highest possible stomatal aperture ($\alpha = 1$) and estimated with stomatal features, had a mean value of $1750 \pm 117 \text{ mmol m}^{-2} \text{s}^{-1}$ and varied from $86 \text{ mmol m}^{-2} \text{s}^{-1}$ to $6214 \text{ mmol m}^{-2} \text{s}^{-1}$ in *Chelidonium majus* and *Malva neglecta* Wallr., respectively. Operational leaf surface conductance ($g_{\text{th op}}$) had a mean value of $1081 \pm 73 \text{ mmol m}^{-2} \text{s}^{-1}$ and varied from $54 \text{ mmol m}^{-2} \text{s}^{-1}$ to $3770 \text{ mmol m}^{-2} \text{s}^{-1}$ in *Chelidonium majus* and *Malva neglecta*,

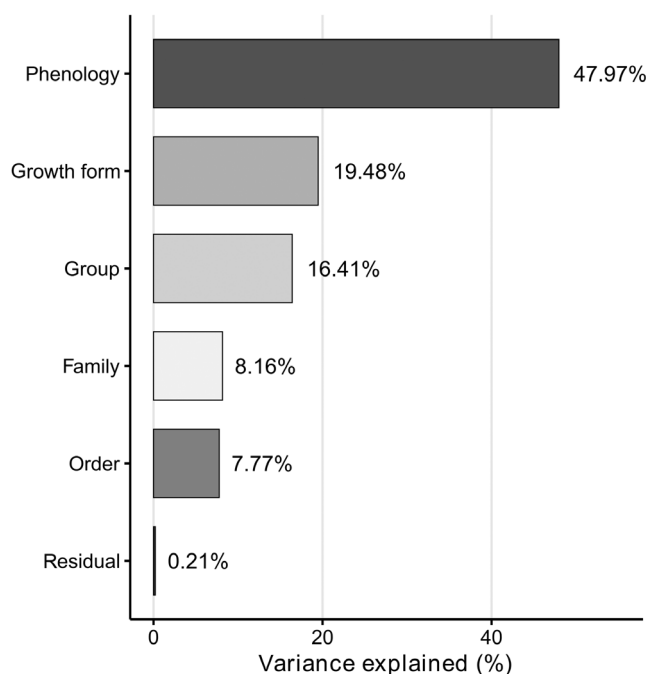


Fig. 4 Variance partitioning of minimum conductance ($g_{\min}$). Bars represent the percentage of total variation explained by each nested level. Levels are ordered by decreasing variance explained. The 'group' level corresponds to the three categories: angiosperms, ferns and allies, and gymnosperms. No. of individual observations = 874.

respectively. All stomatal traits, and modeled $g_{\text{th max}}$ and $g_{\text{th op}}$ conductance rates, had significant phylogenetic signals (Table 1), suggesting phylogenetic conservatism in stomatal anatomy and stomatal-derived conductance rates. In this line, $g_{\text{th max}}$ distribution onto the phylogeny shows that closely related species tend to have similar maximum conductance rates (Figs S5, S6; Table S4).

Coordination between leaf conductance rates, stomatal and leaf functional traits

Both ends of the conductance spectra were not coordinated, as we found a lack of correlation between $g_{\min}$ and $g_{\text{th max}}$ (Fig. 5a; $r = 0.20$; $P = 0.07$), suggesting the lack of a tradeoff between the maximum potential conductance and the measured minimum leaf conductance. $g_{\min}$ and $g_{\text{th op}}$ were marginally but positively correlated (Fig. 5b; $r = 0.22$; $P \leq 0.05$), suggesting a link between theoretical operational conductance, the stomatal pore aperture for photosynthesis, and measured minimum conductance (Table S5). However, the relationship of $g_{\min}$ and $g_{\text{th op}}$ became slightly non-significant when considering phylogenetic relatedness via PICs and PGLS analyses (Tables S6, S7). Both stomatal size and density were not correlated to $g_{\min}$ (Table S5), and this was also observed after considering phylogenetic relatedness (Tables S6, S7). Amongst the studied stomatal traits, SI showed the strongest correlation with $g_{\min}$ (Fig. 5c; $r = 0.34$; $P \leq 0.005$). The evolutionary relationship between $g_{\min}$ and SI was further supported by PGLS analyses (Table S7). Relationships between

the other studied traits were observed across the species level using standard (Table S5) and phylogenetically informed correlations, that is, PICs (Table S6) and PGLS (Table S7). For instance, samples with higher degrees of succulence and higher LMA took significantly longer times to reach $\text{time}_{\text{RWC}80}$ and $\text{time}_{\text{RWC}50}$, yet, both the slope of the dehydration curves and $g_{\min}$ remained unrelated to neither LMA nor leaf succulence. Both $g_{\text{th op}}$ and $g_{\text{th max}}$ were scarcely correlated to the measured leaf traits and showed strong relationships with stomatal traits; however, this was expected since the estimation of both conductance values stem from stomatal features.

A multivariate analysis of trait covariation defined two major axes explaining 63.9% of inertia (Fig. 6). The first axis of the PCA (42.9% of explained variability) was mainly loaded by the conductance rates $g_{\text{th max}}$, $g_{\text{th op}}$, along with $\text{time}_{\text{RWC}80}$ and $\text{time}_{\text{RWC}50}$, and to a lesser degree SD and succulence. The second axis of the PCA (21% of explained variability) was mainly loaded by LMA, SPI , A_{stom} , and A_{leaf} . Overall, $g_{\min}$ had a small contribution to both axes. Increasing $g_{\min}$ was concurrent with higher $g_{\text{th max}}$, A_{leaf} , and SI, while it was opposed to LMA, A_{stom} , and succulence. Phenology types were dispersed along both major axes (Fig. 6). Woody evergreen species were mainly isolated due to their higher LMA, lower $g_{\min}$, and A_{leaf} (Fig. 6). While perennial herbs overlapped with the remaining groups, annual herbs and woody deciduous species were segregated on the second axis (Fig. 6) due to differences in $g_{\text{th max}}$, stomatal features, and A_{leaf} .

Relationships of leaf conductance rates with climate variables

Leaf conductance rates were significantly related to climatic variables (Fig. 7) within the range of the climatic original distribution of the studied species. Measured $g_{\min}$, along with the estimated leaf conductance rates $g_{\text{th op}}$ and $g_{\text{th max}}$, all showed significant negative associations with mean annual temperature (MAT), annual precipitation (AP), and precipitation seasonality (PS). Such relationships between leaf conductance rates and climate became mostly insignificant when considering phenologies separately (Fig. S7). Only woody evergreen species presented a significant negative relationship between $g_{\min}$ and MAT, while only perennial herbs presented negative relationships of $g_{\text{th max}}$ with MAT, AP, and PS (Fig. S7). Aridity index, including its extreme distributions, was not correlated with any of the measured stomatal traits or leaf conductance rates (Table S8). Plant species in high temperature habitats tend to have lower minimum and maximum conductance rates (Fig. 6a,d). Among leaf conductance rates, only $g_{\min}$ had a significantly negative correlation with thermal extremes, as indicated by the relationship with the maximum temperature of the warmest month (Table S8). Moreover, species from habitats with higher or more seasonal precipitation show lower rates of minimum and maximum conductances (Fig. 7b,c,e,f). Overall, stomatal traits were poorly correlated with climate; only SI and SD were negatively related to AP, and SD was negatively related to PS (Table S8).

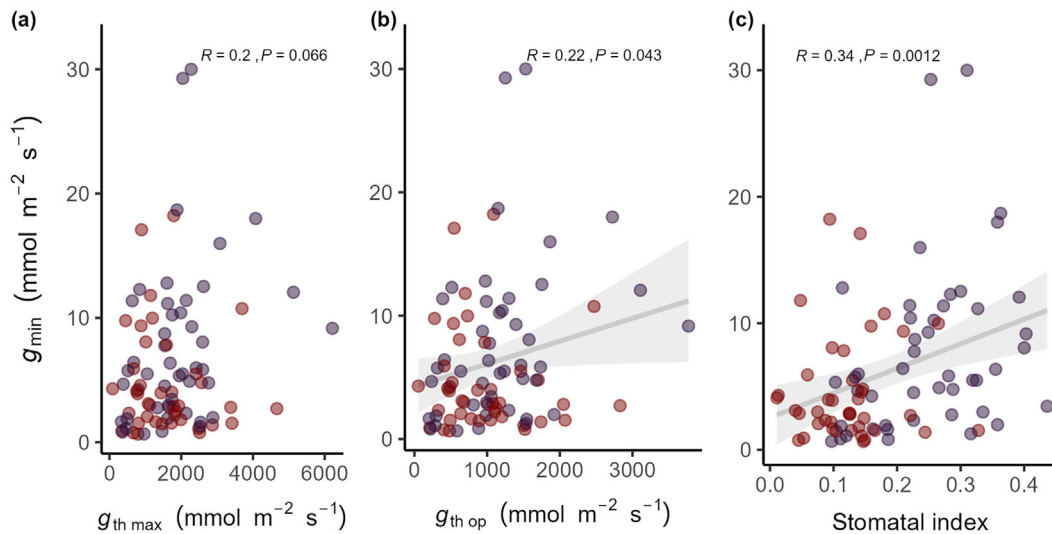


Fig. 5 Minimum conductance ($g_{\min}$) as a function of stomata-derived metrics in vascular plants. Relationships of measured $g_{\min}$ with the leaf maximum achievable ($g_{\text{th max}}$; a) and operational ($g_{\text{th op}}$; b) conductances along with stomatal index (c), derived from stomatal and epidermal traits. Dots indicate amphistomatous (blue) and hypostomatous (red) species. Solid regression lines depict significant correlations; 95% confidence intervals are included for significant correlations.

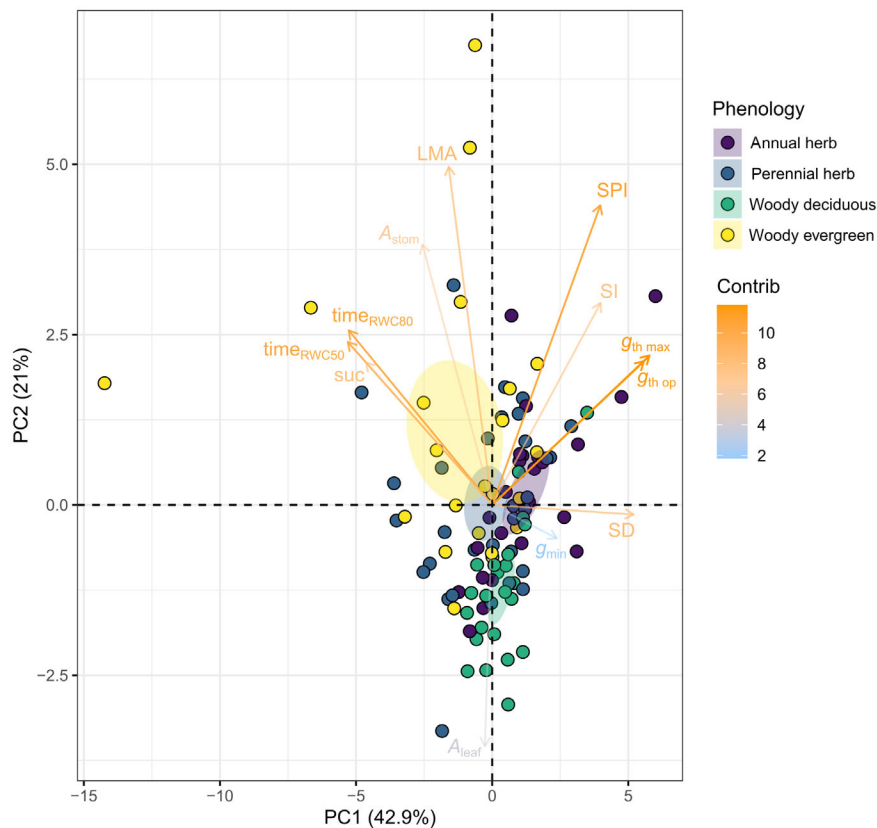


Fig. 6 Principal component analysis (PCA) of minimum and maximum conductance along with leaf and stomatal traits of vascular plants. Arrows and trait color gradients indicate the contribution of each variable to the axes. Species belonging to four different phenology types are indicated in different colors, and 95% confidence ellipses are included. A_{leaf} , leaf area; A_{stom} , stomatal surface; $g_{\text{th max}}$, maximum theoretical leaf conductance; $g_{\min}$, minimum leaf conductance; $g_{\text{th op}}$, operational theoretical leaf conductance; LMA, leaf mass per area; SD, stomatal density; SI, stomatal index; SPI, stomatal pore index; suc, leaf succulence; timeRWC50, time to reach 50% RWC; timeRWC80, time to reach 80% RWC.

Discussion

We found massive variation and strong evidence for convergent evolution in minimum leaf conductance ($g_{\min}$) across 101 species of vascular plants. By using a standardized methodology and

measuring water loss rates between RWC boundaries (80%–50% RWC) following recent recommendations (Burllett *et al.*, 2025), this study represents the largest assessment of the interspecific variability in $g_{\min}$ to date using a uniform method. Further, our research expanded data to underrepresented clades; for instance,

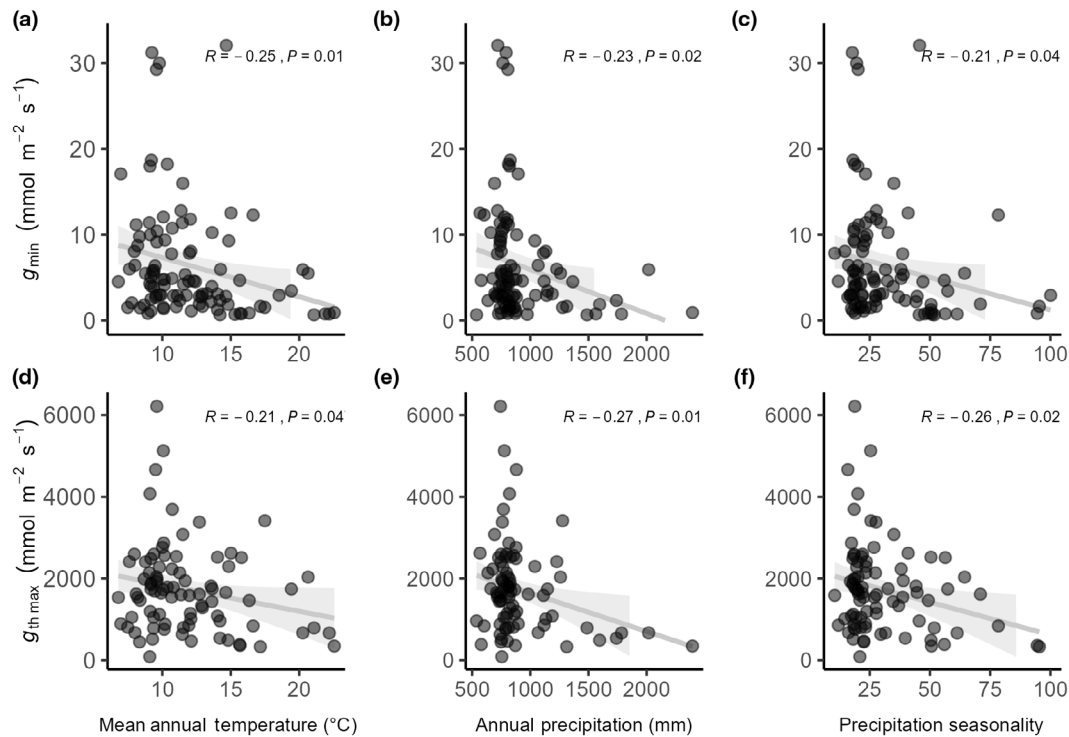


Fig. 7 Relationships between leaf water conductances and species climatic distribution. Measured minimum leaf conductance ($g_{\min}$) as a function of mean annual temperature (a), annual precipitation (b), and precipitation seasonality (c). Estimated maximum leaf conductance ($g_{th\ max}$) as a function of mean annual temperature (d), annual precipitation (e), and precipitation seasonality (f). Solid regression lines depict significant correlations; 95% confidence intervals are included.

the last extensive $g_{\min}$ meta-analysis of Duursma *et al.* (2019) included 221 species, with 136 angiosperms, 49 gymnosperms, but only one pteridophyte. Our analysis of the phylogenetic distribution of $g_{\min}$ values across vascular plants, and the report of a low phylogenetic signal, adds to a large number of hydraulic traits for which the phylogenetic signal has been previously studied (Ávila-Lovera *et al.*, 2023).

The lack of a phylogenetic signal in $g_{\min}$ suggests that minimum water loss values result from convergent evolution, as phylogenetically distant species present similar $g_{\min}$ rates. Although the convergent evolution in $g_{\min}$ could result from other non-apparent reasons (Losos, 2011), it suggests that low minimum leaf conductance values might stem from evolutionary adaptation to avoid water loss. Phylogenetic conservatism seems to be highly variable in plant hydraulic traits. For instance, previous studies have shown a significant phylogenetic signal in P_{50} and an absent, or weak, phylogenetic signal in turgor loss point (TLP) (Guillemot *et al.*, 2022; Ávila-Lovera *et al.*, 2023). We provide compelling evidence of a significant phylogenetic signal in both the stomatal-derived operational ($g_{th\ op}$) and maximum ($g_{th\ max}$) conductance rates. Previous studies have proposed an absence of phylogenetic signal in both stomatal-derived or measured $g_{\max}$ (Fu *et al.*, 2012; Li *et al.*, 2015; Liu & Osborne, 2015). However, such studies analyzed patterns of trait distribution within a more restricted phylogenetic clade or fewer habitats. Our observations of a phylogenetic signal in $g_{th\ max}$ across vascular plants converge with a meta-analysis on 2793 species that

evidenced phylogenetic signal in stomatal conductance rates g_s and $g_{\max}$ (Ávila-Lovera *et al.*, 2023). Given the intrinsic relationship of stomatal traits and stomatal-derived estimations of conductance, the phylogenetic conservatism in $g_{th\ op}$ and $g_{th\ max}$ might arise from the high phylogenetic signal of stomatal traits shown in this study and supported by previous research (Fu *et al.*, 2012; Li *et al.*, 2015).

Plant phenology (i.e. biological life cycles *sensu largo*) might exert an important coercion on plant water economy traits. In this sense we can expect to observe enhanced water economy in plants with longer living leaves such as evergreen trees, which maintain their leaves throughout several years, and perennial herbs, which persist along several reproductive cycles. Previous studies have shown that there is a trend for evergreen leaves to have lower $g_{\min}$ than deciduous leaves (Burghardt & Riederer, 2003; Machado *et al.*, 2021; Slot *et al.*, 2021; Burlett *et al.*, 2025; Sunny *et al.*, 2025). A similar trend of higher rates of water loss in deciduous species has been shown when studying the cuticular transpiration of stomatous, isolated leaf cuticles of 24 species from various biomes (Schreiber & Riederer, 1996). However, a meta-analysis on 160 species concluded a lack of significant differences in $g_{\min}$ by plant growth form as well as between woody deciduous and evergreen species (Schuster *et al.*, 2017). Yet, our study shows clear differences in $g_{\min}$ values between both groups and we show that phenology explained most of minimum conductance variability, with woody deciduous species having higher rates of water loss than evergreen

species. We further extend these observations by showing that annual herbs have significantly higher rates of $g_{\min}$ than perennial herbs. A plausible reason to observe a lack of differences in $g_{\min}$ across growth forms or leaf types using meta-analytic approaches (Schuster *et al.*, 2017; Duursma *et al.*, 2019) might stem from the high variability of $g_{\min}$ across dehydration as well as a consequence of the environmental parameters (T and RH) used during the experiments (Burllett *et al.*, 2025). A meta-analysis on 601 tropical woody species found no differences in drought tolerance traits such as hydraulic vulnerability (P_{50}) or TLP between evergreen or deciduous leaf habits (Guillemot *et al.*, 2022). Our findings suggest that, while hydraulic dysfunction thresholds might be similar across phenology types, they present significant differences in minimum leaf conductance after stomatal closure. This might confer reduced dehydration rates during drought and contribute to achieving longer leaf life spans in evergreen woody species and perennial herbs, which have higher structural investments in their tissues relative to their counterparts (Reich, 2014).

A major relevance of $g_{\min}$ estimation underlies its ability to inform the depletion rates of the water reservoirs, and its determination on the fitness of the plant. In this context, previous studies have shown that minimum leaf conductance variability has a strong effect on the estimation of the time to hydraulic failure, that is the elapsed time between the onset of drought stress and the occurrence of water potential thresholds inducing irreversible hydraulic dysfunction (Machado *et al.*, 2021; Burllett *et al.*, 2025). In this sense, a recent study on saplings of 12 tropical tree species has shown that $g_{\min}$ outstanced other hydraulic traits as a main predictor of time to hydraulic failure and plant survival (Ziegler *et al.*, 2024). Similarly, it has been recently stressed that $g_{\min}$ is a key determinant of the time to reach RWC levels inducing physiological damage (Sunny *et al.*, 2025; Zailaa *et al.*, 2025). In this context, we show a very strong relationship between minimum leaf conductance and $\text{time}_{\text{RCW80}}$ and $\text{time}_{\text{RWC50}}$, highlighting the importance of $g_{\min}$ as a conductance rate that sets the tempo to consume the hydraulic safety margin and reach lethal levels of dehydration (Martin-StPaul *et al.*, 2017; Petek-Petrik *et al.*, 2023). However, the minimum leaf conductance measurements presented here cannot be assumed as an unequivocal surrogate of plant survival under drought since they were measured on detached leaves at similar environmental conditions. In this line, it has been recently shown in a study of 21 tropical species that, amongst nine different traits, $g_{\min}$ had the largest environment-driven variation, as compared to individual- or species-related variation (Boisseaux *et al.*, 2025).

This study uses a unified protocol on species growing under relatively similar conditions, thereby eliminating a potential effect of sample provenance. However, a major limitation is the potentiality of $g_{\min}$ plasticity induced by temperature, amongst other environmental cues. A large body of research has shown that temperature is a main driver of $g_{\min}$ variation (Kerstiens, 2006; Drake *et al.*, 2018; Duursma *et al.*, 2019; Billon *et al.*, 2020; Slot *et al.*, 2021; Zailaa *et al.*, 2024; Fernandes *et al.*, 2025; Garen & Michaletz, 2025), and it has been shown that leaf cuticular permeability has a strong increase at *c.* 40°C

(Schreiber, 2001; Schuster *et al.*, 2016). Other potential sources of variation in $g_{\min}$ can stem from leaf age, deposition of cutin and epicuticular waxes, environmental humidity and seasonal adjustments (Kerstiens, 1996, 2006; Schreiber *et al.*, 2001; Tredenick & Farquhar, 2021; Wang *et al.*, 2024; McAdam *et al.*, 2025). Yet, previous studies on a poplar clone and barley have shown drought-promoted wax and cutin depositions without affecting cuticular transpiration rates (Grünhofer *et al.*, 2022; Shellakkutti *et al.*, 2022). Intraspecific variability in $g_{\min}$ might confer important physiological adaptations but further studies would be needed to understand trait plasticity resulting from environmental pressure under natural growth conditions. Such plasticity could explain the proportion of $g_{\min}$ variability that was not explained here by phenology, growth form or phylogeny.

This study shows important relationships between leaf conductance rates and species environmental distributions. Few studies have explored the link between $g_{\min}$ and climatic variables, proposing that species with low $g_{\min}$ are native to drier environments (Brodribb *et al.*, 2014), and that, across four contrasted biomes, temperate species have higher water losses (Schreiber & Riederer, 1996). Amongst the seven climate variables analyzed here, $g_{\min}$ showed the strongest correlation with the mean annual temperature, suggesting that species from hotter environments have lower rates of $g_{\min}$. Such results might be linked with recent findings showing an overall positive association between cuticle thickness and temperature (Li *et al.*, 2025), although the relationship between $g_{\min}$ and cuticle thickness has not been fully established (Machado *et al.*, 2021). Interestingly, a study on the treeline limit of *Pinus uncinata* supports this trend at the intraspecific level, showing higher $g_{\min}$ and lower cuticle thickness in trees at the higher limit of the species distribution growing in colder conditions (Bueno *et al.*, 2022). Leaf economics might also operate within the relationships between climate and leaf conductance rates, as we observe that plants with more longevous leaves such as perennial herbs and evergreen trees have stronger relationships with climate. Counterintuitively, $g_{\min}$ was negatively related to annual precipitation; yet, it also showed a negative relationship with precipitation seasonality, indicating that species in more seasonal habitats have lower minimum conductance rates. Such $g_{\min}$ declines with increasing precipitation seasonality expand to leaf residual water losses previous results established for other cardinal hydraulic traits (Liu *et al.*, 2021; Vargas *et al.*, 2022), further corroborating the hypothesis of an evolutionary convergence in drought survival strategies. Mean annual temperature, annual precipitation, and precipitation seasonality were also the only climate variables related to $g_{\text{th op}}$ and $g_{\text{th max}}$, suggesting the importance of these climate variables in forcing the selection of different leaf conductance rates.

Seminal works have established leaf, wood, and whole-plant economic spectra based on the covariation of functional and structural traits, which set tradeoffs between construction costs and growth potential (Wright *et al.*, 2004; Chave *et al.*, 2009; Reich, 2014; Díaz *et al.*, 2016). Our study contributes to understanding the integration of minimum, operational, and maximum leaf conductance rates within such economic spectrum. Although $g_{\min}$ increased with $g_{\text{th max}}$, suggesting a tradeoff

between resource acquisition and water loss during drought stress, this trend was statistically insignificant. Further, we reveal a weak positive coordination between $g_{th\ op}$ and g_{min} , which became insignificant when considering phylogenetic relatedness. Previous studies analyzing this association have shown contradictory results. While studies have found a positive relationship between g_{min} and g_{max} measured from gas exchange surveys across nine temperate tree species (Wang *et al.*, 2024) and the stomata-derived $g_{th\ max}$ in 30 species from the Brazilian Cerrado (Machado *et al.*, 2021), another study found a negative relationship between g_{min} and stomata-derived $g_{th\ max}$ in 18 tropical species (Sunny *et al.*, 2025). Using a standard protocol on a large number of species, we show that there is a lack of coordination between measured minimum leaf conductance and theoretical maximum conductance, converging with recent studies and meta-analytical efforts highlighting a weak association of g_{min} with either stomatal conductance or $g_{th\ max}$ (Slot *et al.*, 2021; Ochoa *et al.*, 2024). Our findings suggest that, at operational rates, there is a trend of highly productive leaves to be drought sensitive. This is in line with a previous study showing a positive relationship of measured rates of night-time g_{min} and daylight g_{op} in five *Banksia* species (Drake *et al.*, 2013). However, in contrast to the stomatal safety-efficiency tradeoff established up to the point of stomatal closure (Henry *et al.*, 2019), our results show that after stomatal closure, both ends of the conductance spectrum are disconnected, suggesting that species can potentially achieve high acquisition through maximum conductance while maintaining low water losses.

Previous studies have found significant relationships of water loss via g_{min} with stomatal traits, mainly SD (Muchow & Sinclair, 1989; Machado *et al.*, 2021; Sunny *et al.*, 2025) suggesting stomatal leakiness. Yet, controversy persists as other studies describe the lack of a correlation between SD and g_{min} (Smith *et al.*, 2006; Slot *et al.*, 2021; Ochoa *et al.*, 2024). Although SD had a strong positive association with $g_{th\ op}$ and $g_{th\ max}$ in our dataset, we found a disconnection of both SD and A_{stom} with g_{min} . This reaffirms the influence of stomatal traits on $g_{th\ op}$ and $g_{th\ max}$ conductance rates, which have a greater stomatal contribution that becomes negligible when determining g_{min} (Ochoa *et al.*, 2024). We observed a relationship of g_{min} with the SI, reflecting a compromise between stomatal and epidermal cells (Sack & Buckley, 2016). This relationship suggests that while there is a dissociation of single stomatal traits with minimum leaf conductance, important predictions of g_{min} can emerge when considering the surfaces of epidermal cells. Such a relation between SI and g_{min} invites future research to characterize epidermal pavement cells, which are often neglected, along with the more commonly measured stomatal features. The present study provides information on the phylogenetic organization of leaf conductance rates across 101 vascular plant species and clarifies conductance variability across clades and phenology types. The analysis of phylogenetic signals in leaf conductance features and the evolutionary coordination between associated traits is key in understanding potential tradeoffs that can greatly influence plant fitness. Such understanding might also help in current efforts of model parameterization to extrapolate missing trait values as

inputs for mechanistic models attempting to estimate plant drought vulnerability under ongoing global changes.

Acknowledgements

We thank Gaëlle Capdeville for assistance during the experimental period. We thank Marie-Lise Benot and Jennifer Dudit for assistance in graminoid species identification. All measurements were conducted at the PHENOBOIS Bordeaux platform. This study received financial support from the French government in the framework of the IdEX Bordeaux University 'Investments for the Future' program/GPR Bordeaux Plant Sciences. Open access publication funding provided by COUPERIN CY26.

Competing interests

None declared.

Author contributions

ST, RB and SD designed the study; ST, RB, TB, DN'D and CZ carried out experiments and processed the samples; XB provided tools for data processing; ST, RB, GF and ML performed data analyses; ST wrote the first version of the manuscript; all authors contributed to manuscript revisions.

ORCID

Xavier Bouteiller  <https://orcid.org/0000-0001-8621-383X>
Régis Burtlett  <https://orcid.org/0000-0001-8289-5757>
Sylvain Delzon  <https://orcid.org/0000-0003-3442-1711>
Guillaume Forget  <https://orcid.org/0009-0000-4778-3194>
Maximilien Larter  <https://orcid.org/0000-0001-5730-3203>
Daniela N'Do  <https://orcid.org/0009-0006-8732-9598>
Santiago Trueba  <https://orcid.org/0000-0001-8218-957X>
Camille Ziegler  <https://orcid.org/0000-0002-0855-1347>

Data availability

Codes developed for data acquisition ('cuticular' for Windows) and computation of raw residual conductance ('*gminComputation*' developed as a console version in python, and '*g_Residual*', a script written in R language) are available in the public repository: <https://gitub.u-bordeaux.fr/phenobois>. Data used for this research article is available in the dataverse <https://recherche.data.gouv.fr>; doi: 10.57745/1GIJTS.

References

- Allen CD, Macalady AK, Chenchouni H, Bachelet D, McDowell N, Venetier M, Kitzberger T, Rigling A, Breshears DD, Hogg EH *et al.* 2010. A global overview of drought and heat-induced tree mortality reveals emerging climate change risks for forests. *Forest Ecology and Management* 259: 660–684.
- Ávila-Lovera E, Winter K, Goldsmith GR. 2023. Evidence for phylogenetic signal and correlated evolution in plant–water relation traits. *New Phytologist* 237: 392–407.

- Billon LM, Blackman CJ, Cochard H, Badel E, Hitmi A, Cartiailler J, Souchal R, Torres-Ruiz JM. 2020. The DroughtBox: A new tool for phenotyping residual branch conductance and its temperature dependence during drought. *Plant, Cell & Environment* 43: 1584–1594.
- Blomberg SP, Garland T, Ives AR. 2003. Testing for phylogenetic signal in comparative data: behavioral traits are more labile. *Evolution* 57: 717–745.
- Boisseaux M, Nemetschek D, Baraloto C, Burban B, Casado-Garcia A, Cazal J, Clément J, Derroire G, Fortunel C, Goret J-Y *et al.* 2025. Shifting trait coordination along a soil-moisture-nutrient gradient in tropical forests. *Functional Ecology* 39: 21–37.
- Brodrribb TJ, Holbrook NM, Edwards EJ, Gutiérrez MV. 2003. Relations between stomatal closure, leaf turgor and xylem vulnerability in eight tropical dry forest trees. *Plant, Cell & Environment* 26: 443–450.
- Brodrribb TJ, McAdam SA. 2011. Passive Origins of Stomatal Control in Vascular Plants. *Science* 331: 582.
- Brodrribb TJ, McAdam SAM. 2017. Evolution of the stomatal regulation of plant water content. *Plant Physiology* 174: 639–649.
- Brodrribb TJ, McAdam SAM, Jordan GJ, Martins SCV. 2014. Conifer species adapt to low-rainfall climates by following one of two divergent pathways. *Proceedings of the National Academy of Sciences, USA* 111: 14489–14493.
- Bueno A, Alfathan A, Arand K, Burghardt M, Deininger A-C, Hedrich R, Leide J, Seufert P, Staiger S, Riederer M. 2019. Effects of temperature on the cuticular transpiration barrier of two desert plants with water-spenders and water-saver strategies. *Journal of Experimental Botany* 70: 1613–1625.
- Bueno A, Alonso-Forn D, Peguero-Pina JJ, de Souza AX, Ferrio JP, Sancho-Knapik D, Gil-Pelegrín E. 2022. Minimum leaf conductance (g_{min}) is higher in the treeline of *Pinus uncinata* Ram. in the Pyrenees: Michaelis' hypothesis revisited. *Frontiers in Plant Science* 12: 786933.
- Burghardt M, Riederer M. 2003. Ecophysiological relevance of cuticular transpiration of deciduous and evergreen plants in relation to stomatal closure and leaf water potential*. *Journal of Experimental Botany* 54: 1941–1949.
- Bürkner P-C. 2017. brms: an R package for Bayesian multilevel models using Stan. *Journal of Statistical Software* 80: 1–28.
- Burlett R, Trueba S, Bouteiller XP, Forget G, Torres-Ruiz JM, Martin-StPaul NK, Parise C, Cochard H, Delzon S. 2025. Minimum leaf conductance during drought: unravelling its variability and impact on plant survival. *New Phytologist* 246: 1001–1014.
- Carpenter B, Gelman A, Hoffman MD, Lee D, Goodrich B, Betancourt M, Brubaker M, Guo J, Li P, Riddell A. 2017. Stan: a probabilistic programming language. *Journal of Statistical Software* 76: 1–32.
- Chave J, Coomes D, Jansen S, Lewis SL, Swenson NG, Zanne AE. 2009. Towards a worldwide wood economics spectrum. *Ecology Letters* 12: 351–366.
- Cheng J, Schloerke B, Karambelkar B, Xie Y. 2024. leaflet: create interactive web maps with the JavaScript 'Leaflet' library.
- Cochard H, Pimont F, Ruffault J, Martin-StPaul N. 2021. SurEau: a mechanistic model of plant water relations under extreme drought. *Annals of Forest Science* 78: 55.
- Delf EM. 1912. Transpiration in succulent plants. *Annals of Botany* 26: 409–442.
- Díaz S, Kattge J, Cornelissen JHC, Wright IJ, Lavorel S, Dray S, Reu B, Kleyer M, Wirth C, Colin Prentice I *et al.* 2016. The global spectrum of plant form and function. *Nature* 529: 167–171.
- Drake JE, Tjoelker MG, Vårhammar A, Medlyn BE, Reich PB, Leigh A, Pfautsch S, Blackman CJ, López R, Aspinwall MJ *et al.* 2018. Trees tolerate an extreme heatwave via sustained transpirational cooling and increased leaf thermal tolerance. *Global Change Biology* 24: 2390–2402.
- Drake PL, Froend RH, Franks PJ. 2013. Smaller, faster stomata: scaling of stomatal size, rate of response, and stomatal conductance. *Journal of Experimental Botany* 64: 495–505.
- Duursma RA, Blackman CJ, López R, Martin-StPaul NK, Cochard H, Medlyn BE. 2019. On the minimum leaf conductance: its role in models of plant water use, and ecological and environmental controls. *New Phytologist* 221: 693–705.
- Felsenstein J. 1985. Phylogenies and the comparative method. *The American Naturalist* 125: 1–15.
- Fernandes V d AB, Farnese FS, Arantes BR, Fontineles da Silva ML, Silva FG, Torres-Ruiz JM, Slot M, Cochard H, Menezes-Silva PE. 2025. Leaf minimum conductance dynamics during and after heat stress: implications for plant survival under hotter droughts. *Plant Physiology* 197: kiaf026.
- Fick SE, Hijmans RJ. 2017. WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology* 37: 4302–4315.
- Franks PJ, Beerling DJ. 2009. Maximum leaf conductance driven by CO₂ effects on stomatal size and density over geologic time. *Proceedings of the National Academy of Sciences, USA* 106: 10343–10347.
- Franks PJ, Farquhar GD. 2007. The mechanical diversity of stomata and its significance in gas-exchange control. *Plant Physiology* 143: 78–87.
- Freckleton RP, Harvey PH, Pagel M. 2002. Phylogenetic analysis and comparative data: a test and review of evidence. *The American Naturalist* 160: 712–726.
- Fu P-L, Jiang Y-J, Wang A-Y, Brodrribb TJ, Zhang J-L, Zhu S-D, Cao K-F. 2012. Stem hydraulic traits and leaf water-stress tolerance are co-ordinated with the leaf phenology of angiosperm trees in an Asian tropical dry karst forest. *Annals of Botany* 110: 189–199.
- Garen JC, Michaletz ST. 2025. Temperature governs the relative contributions of cuticle and stomata to leaf minimum conductance. *New Phytologist* 245: 1911–1923.
- Gatscha S, Karambelkar B, Schloerke B. 2024. leaflet.extras: extra functionality for 'leaflet' package.
- Gebrechorkos SH, Sheffield J, Vicente-Serrano SM, Funk C, Miralles DG, Peng J, Dyer E, Talib J, Beck HE, Singer MB *et al.* 2025. Warming accelerates global drought severity. *Nature* 642: 628–635.
- Grossiord C, Buckley TN, Cernusak LA, Novick KA, Poulter B, Siegwolf RTW, Sperry JS, McDowell NG. 2020. Plant responses to rising vapor pressure deficit. *New Phytologist* 226: 1550–1566.
- Grünhofer P, Herzig L, Sent S, Zeisler-Diehl VV, Schreiber L. 2022. Increased cuticular wax deposition does not change residual foliar transpiration. *Plant, Cell & Environment* 45: 1157–1171.
- Guillemot J, Martin-StPaul NK, Bulascoschi L, Poorter L, Morin X, Pinho BX, le Maire G, Bittencourt P, Oliveira RS, Bongers F *et al.* 2022. Small and slow is safe: on the drought tolerance of tropical tree species. *Global Change Biology* 28: 2622–2638.
- Hammond WM, Williams AP, Abatzoglou JT, Adams HD, Klein T, López R, Sáenz-Romero C, Hartmann H, Breshears DD, Allen CD. 2022. Global field observations of tree die-off reveal hotter-drought fingerprint for Earth's forests. *Nature Communications* 13: 1761.
- Henry C, John GP, Pan R, Bartlett MK, Fletcher LR, Scoffoni C, Sack L. 2019. A stomatal safety-efficiency trade-off constrains responses to leaf dehydration. *Nature Communications* 10: 3398.
- Hijmans RJ. 2025. terra: spatial data analysis.
- Jin Y, Qian H. 2019. VPhyloMaker: an R package that can generate very large phylogenies for vascular plants. *Ecography* 42: 1353–1359.
- Josse J, Husson F. 2016. missMDA: a package for handling missing values in multivariate data analysis. *Journal of Statistical Software* 70: 1–31.
- Kerstiens G. 1996. Cuticular water permeability and its physiological significance. *Journal of Experimental Botany* 47: 1813–1832.
- Kerstiens G. 2006. Water transport in plant cuticles: an update. *Journal of Experimental Botany* 57: 2493–2499.
- Li L, McCormack ML, Ma C, Kong D, Zhang Q, Chen X, Zeng H, Niinemets Ü, Guo D. 2015. Leaf economics and hydraulic traits are decoupled in five species-rich tropical-subtropical forests. *Ecology Letters* 18: 899–906.
- Li X, Burrows GE, Dong N, Jordan GJ, Kattge J, Lenz TI, Niinemets Ü, Onoda Y, Poorter H, Sack L *et al.* 2025. A thinner jacket for frosty and windy climates? Global patterns in leaf cuticle thickness and its environmental associations. *New Phytologist* 248: 107–124.
- Liu C, Sack L, Li Y, Zhang J, Yu K, Zhang Q, He N, Yu G. 2023. Relationships of stomatal morphology to the environment across plant communities. *Nature Communications* 14: 6629.
- Liu H, Osborne CP. 2015. Water relations traits of C₄ grasses depend on phylogenetic lineage, photosynthetic pathway, and habitat water availability. *Journal of Experimental Botany* 66: 761–773.
- Liu H, Ye Q, Gleason SM, He P, Yin D. 2021. Weak tradeoff between xylem hydraulic efficiency and safety: climatic seasonality matters. *New Phytologist* 229: 1440–1452.
- López J, Way DA, Sadok W. 2021. Systemic effects of rising atmospheric vapor pressure deficit on plant physiology and productivity. *Global Change Biology* 27: 1704–1720.

- López R, Cano FJ, Martin-StPaul NK, Cochard H, Choat B. 2021. Coordination of stem and leaf traits define different strategies to regulate water loss and tolerance ranges to aridity. *New Phytologist* 230: 497–509.
- Losos JB. 2011. Convergence, adaptation, and constraint. *Evolution* 65: 1827–1840.
- Machado R, Loram-Lourenço L, Farnese FS, Alves RDBF, de Sousa LF, Silva FG, Filho SCV, Torres-Ruiz JM, Cochard H, Menezes-Silva PE. 2021. Where do leaf water leaks come from? Trade-offs underlying the variability in minimum conductance across tropical savanna species with contrasting growth strategies. *New Phytologist* 229: 1415–1430.
- Mantova M, Cochard H, Burlett R, Delzon S, King A, Rodriguez-Dominguez CM, Ahmed MA, Trueba S, Torres-Ruiz JM. 2023. On the path from xylem hydraulic failure to downstream cell death. *New Phytologist* 237: 793–806.
- Martin-StPaul N, Delzon S, Cochard H. 2017. Plant resistance to drought depends on timely stomatal closure. *Ecology Letters* 20: 1437–1447.
- McAdam SAM, Baker CA, Kane CN, Zeisler-Diehl V, Manandhar A, Perdomo MCP, Schreiber L. 2025. Tracking the paths of residual conductance during leaf expansion in *Tilia americana* and *Fagus grandifolia*. *Plant Physiology* 199: kiaf150.
- McAdam SAM, Brodribb TJ. 2012. Stomatal innovation and the rise of seed plants. *Ecology Letters* 15: 1–8.
- Messier J, McGill BJ, Lechowicz MJ. 2010. How do traits vary across ecological scales? A case for trait-based ecology. *Ecology Letters* 13: 838–848.
- Muchow RC, Sinclair TR. 1989. Epidermal conductance, stomatal density and stomatal size among genotypes of *Sorghum bicolor* (L.) Moench. *Plant, Cell & Environment* 12: 425–431.
- Münkemüller T, Lavergne S, Bzeznik B, Dray S, Jombart T, Schiffrs K, Thuiller W. 2012. How to measure and test phylogenetic signal. *Methods in Ecology and Evolution* 3: 743–756.
- Ochoa ME, Henry C, John GP, Medeiros CD, Pan R, Scoffoni C, Buckley TN, Sack L. 2024. Pinpointing the causal influences of stomatal anatomy and behavior on minimum, operational, and maximum leaf surface conductance. *Plant Physiology* 196: 51–66.
- Orme D, Freckleton R, Thomas G, Petzoldt T, Fritz S, Isaac N. 2018. caper: comparative analyses of phylogenetics and evolution in R. R package version 1.0.1.
- Pagel M. 1999. Inferring the historical patterns of biological evolution. *Nature* 401: 877–884.
- Paradis E, Claude J, Strimmer K. 2004. APE: analyses of phylogenetics and evolution in R language. *Bioinformatics* 20: 289–290.
- Pebesma E. 2018. Simple features for R: standardized support for spatial vector data. *The R Journal* 10: 439–446.
- Petek-Petrik A, Petrik P, Lamarque LJ, Cochard H, Burlett R, Delzon S. 2023. Drought survival in conifer species is related to the time required to cross the stomatal safety margin. *Journal of Experimental Botany* 74: 6847–6859.
- R Core Team. 2020. *R: a language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing.
- Reich PB. 2014. The world-wide ‘fast–slow’ plant economics spectrum: a traits manifesto. *Journal of Ecology* 102: 275–301.
- Revell LJ. 2012. phytools: an R package for phylogenetic comparative biology (and other things). *Methods in Ecology and Evolution* 3: 217–223.
- Riederer M, Schreiber L. 2001. Protecting against water loss: analysis of the barrier properties of plant cuticles. *Journal of Experimental Botany* 52: 2023–2032.
- Ruffault J, Pimont F, Cochard H, Dupuy J-L, Martin-StPaul N. 2022. SurEau-Ecos v2.0: a trait-based plant hydraulics model for simulations of plant water status and drought-induced mortality at the ecosystem level. *Geoscientific Model Development* 15: 5593–5626.
- Sack L, Buckley TN. 2016. The developmental basis of stomatal density and flux. *Plant Physiology* 171: 2358–2363.
- Samuels L, Kunst L, Jetter R. 2008. Sealing plant surfaces: Cuticular wax formation by epidermal cells. *Annual Review of Plant Biology* 59: 683–707.
- Šantrůček J, Šimánová E, Karbůlková J, Šimková M, Schreiber L. 2004. A new technique for measurement of water permeability of stomatous cuticular membranes isolated from *Hedera helix* leaves. *Journal of Experimental Botany* 55: 1411–1422.
- Schmitt S, Boisseaux M. 2023. Higher local intra- than interspecific variability in water- and carbon-related leaf traits among Neotropical tree species. *Annals of Botany* 131: 801–811.
- Schneider CA, Rasband WS, Eliceiri KW. 2012. NIH Image to IMAGEJ: 25 years of image analysis. *Nature Methods* 9: 671–675.
- Schreiber L. 2001. Effect of temperature on cuticular transpiration of isolated cuticular membranes and leaf discs. *Journal of Experimental Botany* 52: 1893–1900.
- Schreiber L, Riederer M. 1996. Ecophysiology of cuticular transpiration: comparative investigation of cuticular water permeability of plant species from different habitats. *Oecologia* 107: 426–432.
- Schreiber L, Skrabs M, Hartmann K, Diamantopoulos P, Simanova E, Santrucek J. 2001. Effect of humidity on cuticular water permeability of isolated cuticular membranes and leaf disks. *Planta* 214: 274–282.
- Schuster A-C, Burghardt M, Alfharani A, Bueno A, Hedrich R, Leide J, Thomas J, Riederer M. 2016. Effectiveness of cuticular transpiration barriers in a desert plant at controlling water loss at high temperatures. *AoB Plants* 8: plw027.
- Schuster A-C, Burghardt M, Riederer M. 2017. The ecophysiology of leaf cuticular transpiration: are cuticular water permeabilities adapted to ecological conditions? *Journal of Experimental Botany* 68: 5271–5279.
- Shellakkutti N, Thangamani PD, Suresh K, Baales J, Zeisler-Diehl V, Klaus A, Hochholdinger F, Schreiber L, Kreszies T. 2022. Cuticular transpiration is not affected by enhanced wax and cutin amounts in response to osmotic stress in barley. *Physiologia Plantarum* 174: e13735.
- Slot M, Nardwattanawong T, Hernández GG, Bueno A, Riederer M, Winter K. 2021. Large differences in leaf cuticle conductance and its temperature response among 24 tropical tree species from across a rainfall gradient. *New Phytologist* 232: 1618–1631.
- Smith SA, Brown JW. 2018. Constructing a broadly inclusive seed plant phylogeny. *American Journal of Botany* 105: 302–314.
- Smith SE, Fendeneim DM, Halbrook K. 2006. Epidermal conductance as a component of dehydration avoidance in *Digitaria californica* and *Eragrostis lehmanniana*, two perennial desert grasses. *Journal of Arid Environments* 64: 238–250.
- Sunny R, Venu M, Peddiraju B, Chakrabarty S, Barua D. 2025. It's about time! Leaf minimum conductance determines time to reach critical thresholds for leaf dehydration in a seasonal tropical forest. *bioRxiv*.
- Trabucco A, Zomer R. 2018. Global aridity index and potential evapotranspiration (ET0) climate database v2.
- Tredenick EC, Farquhar GD. 2021. Dynamics of moisture diffusion and adsorption in plant cuticles including the role of cellulose. *Nature Communications* 12: 5042.
- Trueba S, Muñoz NG, Burlett R, Lamarque LJ, Gibon Y, Gimeno TE, Kaisermann A, Benard C, Lemaire C, Torres-Ruiz JM *et al.* 2024. The rates of starch depletion and hydraulic failure both play a role in drought-induced seedling mortality. *Annals of Forest Science* 81: 27.
- Trueba S, Pan R, Scoffoni C, John GP, Davis SD, Sack L. 2019. Thresholds for leaf damage due to dehydration: declines of hydraulic function, stomatal conductance and cellular integrity precede those for photochemistry. *New Phytologist* 223: 134–149.
- Vargas G, Kunert N, Hammond WM, Berry ZC, Werden LK, Smith-Martin CM, Wolfe BT, Toro L, Mondragón-Botero A, Pinto-Ledezma JN. 2022. Leaf habit affects the distribution of drought sensitivity but not water transport efficiency in the tropics. *Ecology Letters* 25: 2637–2650.
- Wang S, Hoch G, Grun G, Kahmen A. 2024. Water loss after stomatal closure: quantifying leaf minimum conductance and minimal water use in nine temperate European tree species during a severe drought. *Tree Physiology* 44: tpa027.
- Wong SC, Cowan IR, Farquhar GD. 1979. Stomatal conductance correlates with photosynthetic capacity. *Nature* 282: 424–426.
- Wright IJ, Reich PB, Westoby M, Ackerly DD, Baruch Z, Bongers F, Cavender-Bares J, Chapin T, Cornelissen JH, Diemer M. 2004. The worldwide leaf economics spectrum. *Nature* 428: 821–827.
- Yang Y-J, Bi M-H, Nie Z-F, Jiang H, Liu X-D, Fang X-W, Brodribb TJ. 2021. Evolution of stomatal closure to optimize water-use efficiency in response to dehydration in ferns and seed plants. *New Phytologist* 230: 2001–2010.
- Zailaa J, Scoffoni C, Brodersen CR. 2024. Stomatal closure as a driver of minimum leaf conductance declines at high temperature and vapor pressure deficit in *Quercus*. *Plant Physiology* 197: kiae551.

- Zailaa J, Trueba S, Browne M, Fletcher LR, Buckley TN, Brodersen CR, Scoffoni C, Sack L. 2025. Sensitive hydraulic and stomatal decline in extreme drought tolerant species of California *Ceanothus*. *Plant, Cell & Environment* 48: 1555–1573.
- Zanne AE, Tank DC, Cornwell WK, Eastman JM, Smith SA, FitzJohn RG, McGlinn DJ, O'Meara BC, Moles AT, Reich PB. 2013. Three keys to the radiation of angiosperms into freezing environments. *Nature* 506: 89–92.
- Ziegler C, Cochard H, Stahl C, Foltzer L, Gérard B, Goret J-Y, Heuret P, Levionnois S, Maillard P, Bonal D *et al.* 2024. Residual water losses mediate the trade-off between growth and drought survival across saplings of 12 tropical rainforest tree species with contrasting hydraulic strategies. *Journal of Experimental Botany* 75: 4128–4147.
- Zizka A, Silvestro D, Andermann T, Azevedo J, Duarte Ritter C, Edler D, Farooq H, Herdean A, Ariza M, Scharn R *et al.* 2019. CoordinateCleaner: standardized cleaning of occurrence records from biological collection databases. *Methods in Ecology and Evolution* 10: 744–751.

Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Variability of minimum conductance ($g_{\min}$) across vascular plant species.

Fig. S2 Phylogeny of studied vascular plants with annotated nodes.

Fig. S3 Comparisons of minimum conductance ($g_{\min}$) values across phenologies in angiosperm and gymnosperm woody species.

Fig. S4 Variance partitioning of minimum conductance ($g_{\min}$) including phylogenetic signal.

Fig. S5 Maximum theoretical conductance ($g_{\text{th max}}$) across vascular plant species.

Fig. S6 Phylogeny of studied vascular plants for stomatal analyses with annotated nodes.

Fig. S7 Relationships between leaf water conductances and species climatic distribution for different phenology types.

Methods S1 Maximum theoretical leaf conductance estimations.

Table S1 Ancestral character estimates of minimum conductance after stomatal closure ($g_{\min}$) across vascular plant species.

Table S2 Average and comparisons of conductance rates, stomatal and leaf functional traits across major clades of vascular plants.

Table S3 Average and comparisons of conductance rates, stomatal and leaf functional traits across phenology types in vascular plants.

Table S4 Ancestral character estimates of theoretical maximum conductance ($g_{\text{th max}}$) across vascular plant species.

Table S5 Correlations between conductance rates, leaf and stomatal traits.

Table S6 Phylogenetic independent contrast correlations between conductance rates, leaf and stomatal traits.

Table S7 Variation in minimum, operational and maximum leaf conductance as explained by leaf conductance rates, leaf functional and stomatal traits.

Table S8 Relationships between climatic variables and stomatal and leaf conductance rates.

Please note: Wiley is not responsible for the content or functionality of any Supporting Information supplied by the authors. Any queries (other than missing material) should be directed to the *New Phytologist* Central Office.

Disclaimer: The New Phytologist Foundation remains neutral with regard to jurisdictional claims in maps and in any institutional affiliations.